

Approximating Irrational Numbers

Worksheet 7A.02.04

Name _____

Date _____

Period _____

1. Find the value of each. If there is no real value, write “no real value.”

$\sqrt{16}$	$\sqrt{-16}$	$\sqrt[3]{27}$	$\sqrt[3]{-27}$
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2. Which of the four in question 1 had no real value? Explain when that happens.

3. Select ALL of the numbers below that are irrational.

☐ $\sqrt{25}$

☐ $\sqrt{18}$

☐ $\sqrt[3]{27}$

☐ $\sqrt[3]{30}$

☐ π

☐ $\frac{4}{9}$

Questions 4 and 5: name the perfect squares or cubes first, then the root.

4. $\sqrt{23}$

a. 23 lies between the perfect squares _____ and _____.

b. So the root lies between the integers _____ and _____.

c. The root is closer to _____.

5. $\sqrt[3]{55}$

- a. 55 lies between the perfect cubes _____ and _____.
- b. So the root lies between the integers _____ and _____.
- c. The root is closer to _____.

6. Approximate $\sqrt{78}$ to the nearest tenth. Fill in the table, then decide.

Try	Result	How far from 78
$8.8 \times \text{itself}$		
$8.9 \times \text{itself}$		

To the nearest tenth, the value is _____.

7. Approximate $\sqrt[3]{120}$ to the nearest tenth. Fill in the table, then decide.

Try	Result	How far from 120
4.9 cubed		
5 cubed		

To the nearest tenth, the value is _____.

8. Use $\sqrt{78} \approx 8.83$ to approximate each expression to the nearest tenth.

Expression	Approximate Value
$\frac{\sqrt{78}}{4}$	
$\sqrt{78} + 5$	
$-3\sqrt{78}$	

9. Approximate the value of each expression to the nearest tenth.

	Expression	Approximate Value
A	$\sqrt[3]{120}$	
B	$\sqrt[3]{120} - 2$	
C	2π	
D	$1 + \pi$	

10. Plot and label A, B, C and D on the number line.



11. Priya says that because $\sqrt{6} \approx 2.4$, the value of $2 \cdot \sqrt{6}$ is 4.8. Her calculator says 4.9. Who is right, and what went wrong?